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Bioelectrical impedance analysis as an alternative to dual-energy x-ray absorptiometry in the assessment of fat mass and appendicular lean mass in patients with obesity

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ABSTRACT

Objective: Obesity is a challenge for bioelectrical impedance analysis (BIA) estimations of skeletal muscle and fat mass (FM), and none of the equations used for appendicular lean mass (ALM) have been developed for people with obesity. By using different equations and proposing a new equation, this study aimed to assess the estimation of FM and ALM using BIA compared with dual-energy x-ray absorptiometry (DXA) as a reference method in a cohort of people with severe obesity.

Methods: This cross-sectional study compared a multifrequency BIA (TANITA MC-780A) versus DXA for body composition assessment in adult patients with severe obesity (body mass index [BMI] of >35 kg/m^2). Comparisons between measured (DXA) and predicted (BIA) data for FM and ALM were performed using the original proprietary equations of the device and the equations proposed by Kyle, Sergi, and Yamada. Bland-Altman plots were drawn to evaluate the agreement between DXA and BIA, calculating bias and limits of agreement (LOA). Reliability was analyzed using intraclass correlation coefficient (ICC). Stepwise multiple regression analysis was used to derive a new equation to predict ALM in patients with obesity and was validated in a subsample of our cohort.

Results: In this study, 115 patients (72.4% women) with severe obesity (mean BMI of 46.1 [5.2] kg/m^2) were included (mean age 43.5 [8.6] y). FM_{DXA} was 61.4 (10.1) kg, FM_{BIA} was 57.9 (10.3) kg, and ICC was 0.925 ($P <$ 0.001). Bias was −3.4 (4.4) kg (−5.2%), and LOA was −14.0, +7.3 kg. Using the proprietary equations, ALMDXA was 21.8 (4.7) kg and ALMBIA was 29.0 (6.8) kg with an ICC 0.868, bias +7.3 (4.0) kg (+34.1%) and LOA −0.5, +15.1. When applying other equations for ALM, the ICC for Sergi, et al. was 0.880, the ICC for Kyle, et al. was 0.891, and the best ICC estimation for Yamada, et al. was 0.914 ($P < 0.001$). Bias was +2.8 (2.8), +4.1 (2.9), and +2.7 (2.8) kg, respectively. The best-fitting regression equation to predict ALM_{DXA} in our population derived from a development cohort (n = 77) was: ALM = 13.861 + (0.259 x H2/Z) − (0.085 x age) − (3.983 x sex [0 = men; 1 = women]). When applied to our validation cohort (n = 38), the ICC was 0.864, and the bias was the lowest compared with the rest of the equations +0.3 (+0.5) kg (+2.7%) LOA −5.4, +6.0 kg.

Conclusion: BIA using multifrequency BIA in people with obesity is reliable enough for the estimation of FM, with good correlation and low bias to DXA. Regarding the estimation of ALM, BIA showed a good correlation with DXA, although it overestimated ALM, especially when proprietary equations were used. The use of equations developed using the same device improved the prediction, and our new equation showed a low bias for ALM.

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Introduction

Sarcopenic obesity has become a challenge as evidence increasingly shows that low muscle mass and quality have relevant negative implications in people with obesity; in addition to frailty and disability, sarcopenic obesity may lead to metabolic impairment and increased morbidity and mortality. The European Society for Clinical Nutrition and Metabolism (ESPEN) and the European Association for the Study of Obesity (EASO) have recently recognized obesity with altered body composition due to low skeletal muscle function and mass (sarcopenic obesity) as a scientific and clinical priority for researchers and clinicians [1]. However, a clear definition and diagnostic criteria for this clinical entity are still lacking. A recent systematic review has brought to light the differences in the definitions of obesity and sarcopenia, in the methodologies used to assess body composition and physical function, and in the reference values for diagnosing this entity [2]. Specifically, 19 different measurements of low muscle mass and 10 measurements of adiposity were applied across the studies reviewed, with appendicular lean mass (ALM) divided by weight (ALM/wt) or adjusted by height in meters squared (ALM/h^2) being the most frequently applied measurements of sarcopenia and BMI for obesity [2]. Even more, measuring skeletal muscle and fat mass (FM) is not an easy task in clinical practice. Although magnetic resonance imaging, computed tomography (CT) scans, or dual-energy x-ray absorptiometry (DXA) have been considered valid reference methods, they are not easily available in clinical practice, and CT and DXA may involve x-ray exposure [1]. DXA should be performed no more than twice per year, which limits its use for body composition measurement [3].

Bioelectrical impedance analysis (BIA) has been proposed as an alternative tool in the absence of confounding fluid balance abnormalities as it is portable, easy to use, relatively low cost, and free from radiation [4–6]. The BIA analysis method is based on the application of an alternating low-intensity electrical current through biological structures, which produces an opposition or impedance (Z) to the flow of this current [7]. Body water, because of the presence of electrolytes, transmits electrical currents, while fat and bone tissue are relatively non-conductive. In this way, the impedance would be a measure of the total body water (TBW), and it allows estimation of fat-free mass (FFM) assuming that, in a healthy organism and without alterations of the hydric compartments, the TBW represents a constant fraction of 73.2% of the FFM [5]. Using regression equations by means of the electrical data offered by the BIA allows estimations of skeletal muscle mass and ALM [8–12]. Multifrequency BIA measures the impedance at multiple frequencies (0, 1, 5, 50, 100, 200 to 500 kHz), thus allowing a more accurate estimate of the TBW.

Several studies have strongly supported the value of BIA, especially multifrequency BIA (MF-BIA), in ALM prediction [8,10,11,13–17]. However, BIA does not actually measure either fat or muscle mass, but it estimates using a calibration equation developed using a reference method such as DXA, CT, or magnetic resonance imaging [18]. Each equation has been developed in a different population using different devices, so their applicability should be limited to that specific context as their results may differ when not used in the same population [14]. A recent systematic review has addressed equation models developed with BIA to assess muscle mass. Most of them explained more than 80% of the variance between BIA and a reference method, usually DXA [9], but tended to overestimate muscle mass [17,19].

Obesity is specifically a challenge for BIA estimations, as the assumptions of constant hydration is violated with increasing adiposity because of the higher ratio of extracellular to intracellular water in the adipose tissue–part of connective tissue [20]. Additionally, none of the equations collected in the review by Beaudart, et al. [9] had been developed in people with obesity, although obesity is one of the clinical scenarios in which body composition measurement is more important. In a previous cross-sectional study including 40 patients with severe obesity [21], our group reported a good correlation between single-frequency BIA and DXA, although overestimation of FFM and underestimation of FM was seen. Our current study aimed to assess the estimation of FM and ALM using an MF-BIA compared with DXA as a reference method in a cohort of people with severe obesity (BMI >35 kg/m^2) using different equations. Since none of the equations were developed in people with obesity, our second objective was to propose a new equation for this population.

Materials and methods

This is a cross-sectional study to compare body composition assessment obtained by BIA using DXA as a reference method in patients with obesity. Patients who were assessed in our high-risk obesity clinic between April 2017 and December 2020 were enrolled. Inclusion criteria were patients 18 to 60 y old with a body mass index (BMI) higher than 35 kg/m^2 who were able to walk more than 10 m and able to provide written informed consent. Exclusion criteria were weight more than180 kg, difficulty maintaining an upright position, or presence of pacemakers. The study was approved by the Ethical Research Committee of our institution (approval n° 1775/2017), and all patients provided written informed consent as per the requirements of the Declaration of Helsinki.

Measurements

Height and weight were measured with digital SECA model 701 (SECA Deutschland, Hamburg, Germany) for each participant and recorded to the nearest 0.1 cm and 0.1 kg. Each patient underwent both whole-body DXA scan and BIA at our medical facility, performed on the same day. A Lunar iDXA (GE Healthcare, USA) was used to obtain FFM, FM, and ALM as the sum of the arm and leg lean mass. A standing-position 8-electrode MF-BIA segmental body composition analyzer (MC-780A; TANITA, Tokyo, Japan) was used to measure bioelectrical impedance (Z), resistance (R), and reactance (Xc) at 5, 50, and 250 kHz frequency (Z5, Z50, and Z250). Participants were instructed to avoid vigorous exercise and consuming alcohol for the 24-h period before the experiment, to finish the last meal at least 2 h before the measurement, and to empty their bladder before the measurement. Patients were evaluated in their underwear, in a standing position, and were asked to stand barefoot with toe-and heel electrodes and to hold handgrips with arms hanging down a few centimeters from the hip.

Estimations of FM, FFM, and ALM

Using masked proprietary original prediction equations included in the software of the BIA tool and devised by Dr. Steven Heymsfield and his research team at St Luke's Roosevelt Hospital, Columbia University, New York [22], we obtained estimations for FM, FFM, and ALM. ALM was also estimated using other equations. The most reliable results to estimate ALM in the adult White population [9] were reported by Sergi, et al. [11] in older adults, with a determination coefficient R^2 of 0.923. The equation by Kyle, et al. [8] was selected for being that with a higher R^2 of 0.953 and because it was developed in a European population, like ours, and also because it had a mean age closer to our patients. Also the BMI in that study was below 25 kg/m^2 [8]. Both equations were performed on a SF-BIA, so the equation by Yamada, et al. [10] was also tested as it was developed using the same MF-BIA device as ours, but in a healthy Japanese population. Every comparison between measured (DXA) and predicted (BIA) data for ALM was performed not only using the original proprietary equations of the device but also using these three equations.

ALM_{Kyle}: −4.211+ (0.267 x height^2/R) + (0.095 x weight) + (1.909 x sex (male = 1; female = 0) + (0.012 age) + (0.05 xXc)

ALM_{Sergi}: −3.964+ (0.227 x height ^{2}/R) + (0.095 x weight) + (1.384 x sex (male =1; female =0) + (0.064 xXc)

ALM_{Yamada} = (0.6947 x (height /Z50)) + (−55.24 x [Z250/Z5]) + (−10940 x [1/Z50]) + 51.33 for men, and ALM_{Yamada} = (0.6144 x (height /Z50)) + (−36.61 x [Z250/Z5]) + (−9332 x [1/Z50]) + 37.91 for women

Statistical analysis

Data were analyzed using SPSS software (version 15.0, Chicago, IL, USA). Continuous data are presented as mean (SD) [23], after checking for normality using Kolmogorov-Smirnov test. Categorical data are presented as frequencies and percentages (%). One-sample *t* tests and unpaired Student's *t* tests were performed to identify any differences among various characteristics. A *P* value <0.05 was

Table 1
Anthropometric measures and BIA data

	Total N = 115	Men n = 32	Women n = 83
Age (y)	43.5 (8.6)	42.7 (9.1)	43.8 (8.5)
Weight (kg)	126.2 (17.4)	139.8 (16.3)	121.1 (14.9)†
Height (m)	1.65 (0.1)	1.74 (0.1)	1.62 (0.1)†
BMI (kg/m^2)	46.1 (5.2)	45.8 (5.1)	46.2(5.3)
Waist circumference (cm)	136.9 (15.3)	146.8 (7.3)	132.4 (15.9)*
Calf circumference (cm)	48.9 (3.7)	48.6 (3.2)	48.9 (3.9)
Resistance (50 kHz) (Ω)	494.2 (64.4)	454.5 (60.3)	509.3 (59.6)†
Reactance (50 kHz) (Ω)	51.9 (7.2)	51.7 (7.6)	52.0 (7.1)
Phase angle (°)	6.1 (0.5)	6.5 (0.5)	5.9 (0.5)†

BIA, bioelectrical impedance analysis; BMI, body mass index
Statistical differences between men and women marked as *$P < 0.05$
*$P < 0.05$
†$P < 0.001$

considered significant. Bland-Altman plots were drawn to evaluate any agreement between DXA and BIA [24]. The differences between BIA and DXA against their mean values are plotted for each body composition measurement (FM, FFM, ALM). The bias is the difference between the measured (DXA) and predicted (BIA) values of FM/FFM/ALM, and the error is the standard deviation of the bias. The limits of agreement (LOA), calculated as mean difference (bias) ± 1.96 SD (i.e., 95% confidence interval of the individual difference), were used to test agreement between the two methods (measured and predicted ALM values). Percentage of bias was calculated as the bias divided FM/ALM measured by DXA. Pearson's correlation coefficient was calculated to compare the prediction accuracy between BIA equations and DXA measurements. Reliability analysis was performed using the intraclass correlation coefficient (ICC), considering low concordance if ICC was <0.40, fair/good if ICC was 0.41 to 0.75, and very good if ICC was >0.75 [25].

Development of a new equation

Stepwise multiple regression analysis was applied to derive the best-fitting regression equations to predict ALM_{DXA} in our population of patients with obesity. We split our cohort into one group for development (two-thirds of patients) and one group for validation (one-third of patients), similar to the approach developed by Yamada, et al. [10]. The goodness of fit was assessed by the determination coefficient (R^2, the proportion of the total variance in the dependent variable that is explained by the independent variables). All impedance measurements were adjusted for height as impedance index, height^2/Z (cm^2/ohm) [8], and this impedance index at 50 kHz was used as the primary independent variable, and then secondary variables were tested as additional predictors (R and Xc at 50 kHz, weight, age, BMI). Yamada, et al. [10] indicated that expansion of ECW (extracellular water) relative to ICW or TBW is observed with ageing and obesity. The Z at low-frequency (≤50 kHz, for example, Z5 or Z50) currents mainly reflects ECW, while Z at high-frequency (≥250 kHz) reflects TBW. Thus, the impedance ratio of Z250 to Z5 (Z250/Z5) is an index of ECW/TBW, and we included it as the independent variable to estimate ALM. Additionally, the possibility of edema affects ALM estimation, so they proposed the index of 1/Z5 or 1/Z50 as an adjusting variable for this situation. We also tested these parameters in our regression equations. Optimal combination of predictor variables was selected by considering the correlation between predicted and observed data (which should be maximized).

Results

This study included 115 patients with severe obesity (BMI 36.2–61.8 kg/m$_2$), 72,4% of them being women with a mean age of 43.5 (8.6) y. Anthropometric measures and BIA data, stratified by sex, are reported in Table 1. Men had higher weight, height, and waist circumference as well as higher resistance and phase angle, but age and BMI were similar regardless of sex. FM and ALM measurement with DXA and estimations with BIA using the original proprietary equations of our device are shown in Table 2. Table 3 includes the estimations of FM and ALM with BIA compared with DXA measurements using proprietary equations as well as the different selected equations. FM_{DXA} was 61.4 (10.1) kg, and FM_{BIA} was 57.9 (10.3) kg, which meant ICC was 0.925 ($P < 0.001$) (Tables 2 and 3). Bias between DXA and BIA for FM was −3.4 (4.4) kg and LOA −14.0, +7.3 kg. ALM_{DXA} was 21.8 (4.7) kg, and ALM_{BIA} using the original proprietary equations was 29.0 (6.8) kg; ICC was 0.868; bias was +7.3 (4.0) kg; and LOA was −0.5, +15.1 (Tables 2 and 3). When applying other equations for ALM, ICC was almost perfect for Sergi's equation (0.880) and Kyle's equation (0.891), and ICC was the best for Yamada's equation (0.914 [$P < 0.001$]) in our sample. Bias was +2.8 (2.8), +4.1 (2.9), and +2.7 (2.8) kg, respectively. The percentage of bias is also shown in Table 3 and was smaller for FM (−5.2%) but was very high for ALM using proprietary equations (+34.1%) and was the best for the Yamada's equation (+13.3%). Bland-Altman plots, including bias and LOA for the estimations of FM, MM, and ALM using proprietary equations, are shown in Figure 1.

Stepwise multiple regression analysis was applied to derive the best-fitting regression equations to predict ALM_{DXA} in our population of patients with obesity. There were no significant differences in either sex, age, weight, BMI, FM_{DXA}, or ALM_{DXA} between the two cohorts of development (n = 77) and validation (n = 38) (data not shown). The impedance index height^2/Z at 50 kHz was used as the primary independent variable and accounted for 64.8% of the variance of the prediction of ALM. Including age and sex in the equation meant an increase in adjusted R^2 to 0.760 (standard error estimation 2.357, $P < 0.001$). Including in the equation weight or BMI, or other electrical measures such as Z250/Z5 or 1/Z50, did not improve the regression model, and their coefficients were not significant, so we did not include these variables in the model.

The best-fitting regression equation to predict ALM_{DXA} in our population derived in our development cohort (n = 77) was: ALM = 13.861 + (0.259 x H2/Z) − (0.085 x age) − (3.983 x sex [0 = men; 1 = women]). When we applied the equation to our validation cohort (n = 38), the ICC was also very good (0.864). The bias was the lowest compared with the rest of the equations +0.3 (+0.5) kg (+2.7%), LOA −5.4, +6.0 kg (Table 3).

Discussion

Obesity is one of the most important health problems in the Western world [26]. More recently, sarcopenic obesity has emerged as an additional challenge, and ESPEN and EASO statements recognize it as a scientific and clinical priority for

Table 2
Fat mass, lean mass, and appendicular lean mass measurement with DXA and estimations with MF-BIA TANITA MC 780 MA using original equations

	DXA			BIA (Proprietary Equations)		
	Total	Men	Women	Total	Men	Women
Fat mass (kg)	61.4 (10.1)	60.4 (11.5)	61.7 (9.6)	57.9 (10.3)	59.1 (11.7)	57.4 (9.8)
Fat mass/weight (%)	51.4 (5.9)	45.7 (5.6)	53.5 (4.4)*	45.8 (4.4)	42.0 (4.8)	47.2 (3.3)*
Lean mass (DXA) (kg)/SMM (BIA)	58.5 (10.8)	71.6 (8.5)	53.4 (6.4)*	65.2 (9.9)	77.3 (7.9)	60.6 (5.8)*
Muscle mass/weight (%)	46.3 (5.7)	51.5 (5.3)	44.3 (4.6)*	51.8 (4.0)	55.5 (3.8)	50.3 (3.0)*
ALM (kg)	21.8 (4.7)	27.4 (4.2)	19.7 (2.7)*	29.0 (6.8)	38.4 (5.4)	25.5 (2.5)*
ALM/weight (%)	17.3 (3.0)	19.7 (2.8)	16.4 (2.6)*	22.9 (3.3)	27.5 (2.4)	21.1 (1.2)*

ALM, appendicular lean mass; DXA, dual-energy x-ray absorptiometry; MF-BIA, multifrequency bioelectrical impedance analysis; SMM, skeletal muscle mass
*Men vs women: $P < 0.001$

Table 3
Estimations of fat mass and ALM with different equations compared to DXA measurements

Equation	Fat Mass (kg)									ALM (kg)								
	Proprietary Equation			Proprietary Equation			Sergi			Kyle			Yamada			CAULE (validation cohort) n = 38		
	Total	Men	Women	Total	Men	Women	Total	Men	Women	Total	Men	Women	Total	Men	Women	Total	Men	Women
Correlation (Pearson)	0.860†	0.846†	0.878†	0.821†	0.703†	0.341†	0.797†	0.720†	0.400†	0.805†	0.730†	0.401†	0.848†	0.773†	0.536†	0.764†	0.702	0.414†
Adjusted R^2	0.699	0.707	0.768	0.699	0.473	0.105	0.636	0.502	0.160	0.644	0.518	0.150	0.716	0.583	0.278	0.572	0.493	0.171
Bias, compared to DXA (kg)	(SEE 2.687)† −3.4 (4.4)	(SEE 4.639)† −1.3 (6.4)	(SEE 6.208)† −4.2 (4.8)	(SEE 2.687)† +7.3 (4.0)	(SEE 3.072)† +11.1 (3.9)	(SEE 2.507)† +5.8 (2.9)	(SEE 2.833)† +2.8 (2.8)	(SEE 2.985)† +2.0 (2.9)	(SEE 2.444)† +3.1 (2.8)	(SEE 2.787)† +4.1 (2.9)	(SEE 2.938)† +4.2 (2.9)	(SEE 2.443)† +4.0 (2.9)	(SEE 2.488)† +2.7 (2.8)	(SEE 2.730) +3.6 (3.2)	(SEE 2.252)† +2.4 (2.6)	(SEE 2.881) +0.3 (+0.5)	(SEE 3.369) +1.4 (3.2)	(SEE 2.775)† −0.1 (2.8)
Limits of agreement (kg)	−14.0, +7.3	−13.9, +11.3	−13.6, +5.2	−0.5, +15.1	+3.5, +18.6	+0.1, +11.6	−2.8, +8.31	−3.7, +7.8	−2.3, +8.4	+1.5, +9.7	−1.5, +9.9	−1.6, +9.7	−2.8, +8.2	−2.7, +9.8	−2.7, +7.5	−5.4, +6.0	−4.5, +7.7	−5.5, +5.4
Percentage of bias (%)	−5.2 (9.0)	−1.8 (11.1)	−6.6 (7.8)*	+34.1 (18.0)	+41 (16.4)	+31.3 (17.8)†	+14.7 (15.1)	+8.7 (11.6)	+17.0 (15.7)†	+20.5 (15.8)	+16.7 (12.3)	+ 22.0 (16.7)	+13.3 (13.9)	+13.5 (12.6)	+13.2 (14.4)	+2.7 (14.0)	+6.7 (11.8)	+1.5 (14.6)
Intraclass correlation coefficient	0.925†	0.917†	0.935†	0.868†	0.809†	0.507†	0.880†	0.809†	0.568†	0.891†	0.832†	0.572†	0.914†	0.865†	0.697†	0.864†	0.739†	0.531†

ALM, appendicular lean mass; DXA, dual-energy x-ray absorptiometry; SEE, standard error estimation; CAULE, complejo asistencial universitario de León.
Differences men vs. women: *$P < 0.05$
†$P < 0.001$.

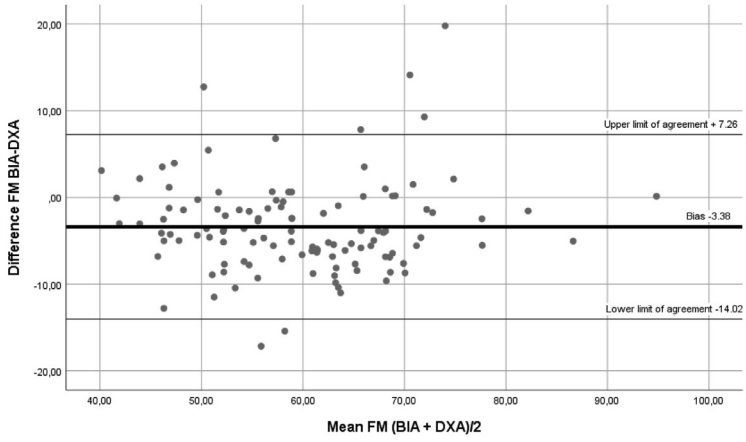
researchers and clinicians [1]. However, diagnosing sarcopenic obesity is not an easy task, as diagnostic criteria are still lacking and measuring skeletal muscle and FM is not easy in clinical practice. Several studies have strongly supported the value of BIA, especially multifrequency BIA, in appendicular skeletal muscle mass prediction [8,10,11,13–17]. Nevertheless, we need to consider that BIA may underestimate FM and overestimate muscle mass [17,19,27,28].

In people with obesity, some assumptions required for BIA are brought into question and, furthermore, prediction equations have not been developed specifically for this population [9,29]. Our group has previously reported a good correlation between single-frequency BIA (TANITA TBF 300) and DXA in people with obesity (R = 0.918 for FM and R = 0.896 for FFM), although overestimation of FFM and underestimation of FM did occur [21].

In the current study, we aimed to assess the estimation of FM and ALM using a MF-BIA compared with DXA as a reference method in a bigger cohort of people with severe obesity, using different equations and proposing a new equation for this population. Our population was young (mean age 43.5 y) but with severe obesity (mean BMI 46.1 kg/m^2), and few studies have addressed a similar group of people despite the relevance of measuring body composition in this context. Using the same BIA as our study, Verney, et al. [30] reported satisfactory FM and FFM measures in healthy young people, compared with DXA, with an ICC of 0.84. ALM was not considered in the study. Considering FM, our results also showed a very good ICC of 0.925 (Table 3), even higher than those studies performed in healthy people, so despite a bias of −3.4 kg (Fig. 1), we could consider this high ICC allows a reliable estimation of FM in people with obesity using a multifrequency BIA, as bias results in 5.2% of the real measure (only 1.8% in men but 6.6% in women). Other studies have obtained similar results in people with obesity [29,31]. Becroft, et al. [32] conducted a systematic review to assess the validity of MF-BIA to measure body composition in patients with obesity, confirming that BIA underestimates FM (−3.83 kg, range −1.90 to −6.60 kg), although with a moderate to strong correlation, ranging from Pearson's coefficient R = 0.81 to 0.96. Our results regarding FM are similar. In summary, there is a systematic error in the estimate, but the good correlation would allow the use of BIA as a method for assessing FM.

However, estimating ALM needs the use of population- and device-specific equations. Some of the equations have been developed and validated only in Asian populations [10,15–17] and others in older adults [11,12,17]. Only one equation, developed for healthy Japanese people, has been established for the BIA device used in our study [10]. A recent systematic review has reviewed the equation models developed with BIA to assess muscle mass [9] and allows verification of the existence of a prediction equation when using an specific BIA device for estimating muscle mass, which better suits our population [17,19]. The best results to estimate ALM in those applied to an adult White population were reported by Sergi, et al. [11] in older adults with a mean BMI of 27.0 kg/m^2. They reported a determination coefficient of R^2 = 0.923 (which means that it explained 92.3% of the variance). Kyle's equation was also quite reliable, with R^2 = 0.953 and had the advantage of having been developed in a population with a mean age closer to our patients, although the BMI in their study was below 25 kg/m^2 [8]. We decided to test these two equations for these good results to assess their applicability in our patients with obesity. Both equations were performed on a single-frequency BIA, so we also tested Yamada's equation [10] as it was developed using the same MF-BIA device as ours, but in a Japanese healthy population, reporting R^2 = 0.851 for men and 0.757 for women. Applying

1a



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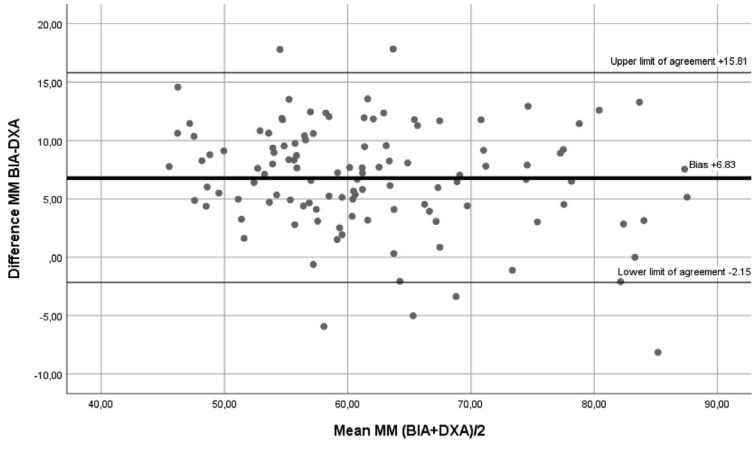

1c

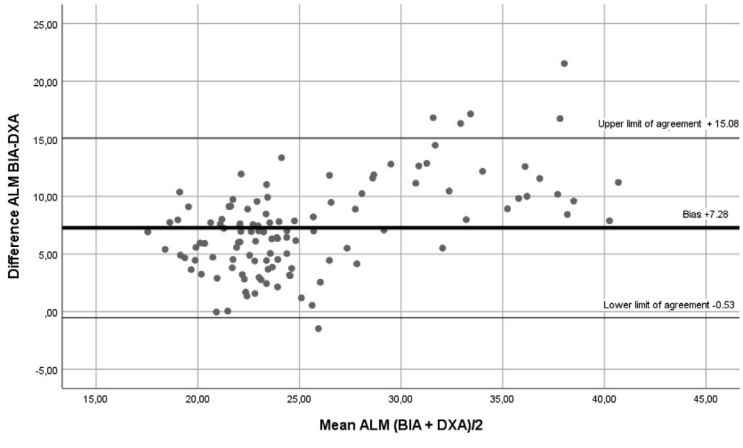

Fig. 1. Bland-Altman plots for the estimations of FM (A), MM (B), and ALM (C) with proprietary equations. ALM, appendicular lean mass; FM, fat mass; MM, muscle mass.

different equations to the same population showed ASM values that correlated strongly with ASM by DXA in a healthy Australian population [14]. In this study, the equation by Sergi, et al. [11] predicted ASM the best and even demonstrated the greatest predictive accuracy in overweight and obese individuals.

In our population, the ICC of 0.868 for the estimation of ALM with our MF-BIA, using the original proprietary equations, was considered very good [25], although it was only good in women (0.507) (Table 3). The results were similar using the equations by Sergi, et al. and Kyle, et al. and were the best with the equation by Yamada, et al. However, the estimation was always worse for women. Yamada, et al. [10] also reported worse results for women. In both their study and ours, the mean age was similar, around 46 y old, and was younger than in other studies [9]. This means that many of the women in their and our studies were of reproductive age. Perhaps hormonal changes that could alter the water component of body composition could explain part of the difference. Unfortunately, we did not control or record the moment of the menstrual cycle of our female patients, and we can consider this one of the limitations of our study.

Despite a good ICC, the estimations of ALM showed a high bias when using the original equations included in the software of the device, especially in men. The best results were obtained again using Yamada's equation [10], which overestimates ALM in 13.3%, without differences between men and women. Therefore, this equation seems to be the most suitable for our population, considering both the ICC and bias. We can consider that this was to be expected since the equation was also developed in a population of similar age and with the same device that we use, although in a Japanese population without obesity.

Single-frequency [8,11,14] or multifrequency [10,15,19] BIA might produce different results in estimating ALM. MF-BIA can estimate extracellular water with greater accuracy, which is important as the ratio of extracellular to intracellular water increases with age and with obesity [10,14]. Therefore, the MF-BIA, like the one used in our study and by Yamada, et al., should be the preferred method for estimating ALM, especially for people with either obesity or advanced age. However, as far as we know, there are currently no prediction equations for ALM that have been developed using MF-BIA devices in people with obesity. This way, one of the strengths of our study is that equations developed with different devices and in different populations have been applied for the first time to a cohort of people with obesity using a multifrequency instrument and establishing a new equation for this people. The equation by Yamada, et al. [10], using a MF-BIA device, proved to be the best in our population. On the other hand, the proprietary equation seems to be the worst. However, the systematic error involved in estimating ALM by BIA using the best equation for our population is 13%, which we must consider.

The last part of our study was to develop a new equation to predict ALM_{DXA} in our population of patients, with obesity using stepwise multiple regression analysis. Although some studies have developed equations for prediction of FFM in obesity [31,33], as far as we know, there is no equation for ALM in this context. When we applied the new equation to our validation cohort, the ICC was also very good (0.864), although no better than obtained by Yamada's equation, and the R^2 was lower. The bias, however, was the lowest compared with the rest of the equations.

The main limitation of our study is the sample size, which included 115 people. As severe obesity is not a frequent condition, we could not anticipate a bigger sample. Additionally, there are few studies regarding this population, making a sample size estimation difficult. Considering that assessing body composition in people with severe obesity is of paramount importance, we would like to offer our study as an exploratory study in a convenience sample to assess if BIA could be considered for this purpose. Perhaps the sample size is the reason for not improving the results given by the equation by Yamada et al., which was developed in 756 individuals. As no equation was good enough in our patients with severe obesity, we developed a new one, but the sample was even smaller as we used our initial sample to both develop and validate a new equation. In their systematic review, Beaudart, et al. [9], recommend that if an equation exists and has been validated by a study free of a high-risk bias, its use is recommended because the development of a new equation in the same context seems redundant. The lack of validation in people with obesity led us to try to develop our own equation, but our sample probably did not have the appropriate size for it, which is a limitation. Therefore, although this study is an attempt to develop an equation for obesity, more studies will be needed to validate it.

In conclusion, our study has shown that BIA using MF-BIA in people with obesity is reliable enough for the estimation of FM. Regarding ALM, BIA is a good alternative considering the high correlation with the measurements made with DXA, although it should always be noted that it overestimates ALM; therefore the cutoff points for people with obesity will need to be established in further studies. Although BIA and DXA cannot be seen as interchangeable, BIA can be considered an easy, non-invasive, and low-cost method for FM and ALM assessment in people with obesity [29], as long as we accept this systematic error associated with the measurements and select an adequate equation. Future research is needed to develop and validate new BIA-specific equations, such as ours, for ALM in patients with different grades of obesity.

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